Engine Model(s): PT6A

73-10-02

FUEL PUMP - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC01-001	Fuel, Engine
PWC03-001	Oil, Engine
PWC05-061	Cloth, Abrasive
PWC05-101	Cloth, Abrasive
PWC05-166	Solution Treatment
PWC05-168	Grit, Vapor Blasting
PWC11-027	Solvent, Cleaning
PWC11-031	Solvent, Cleaning
PWC11-038	Solvent, Cleaning

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Servicing

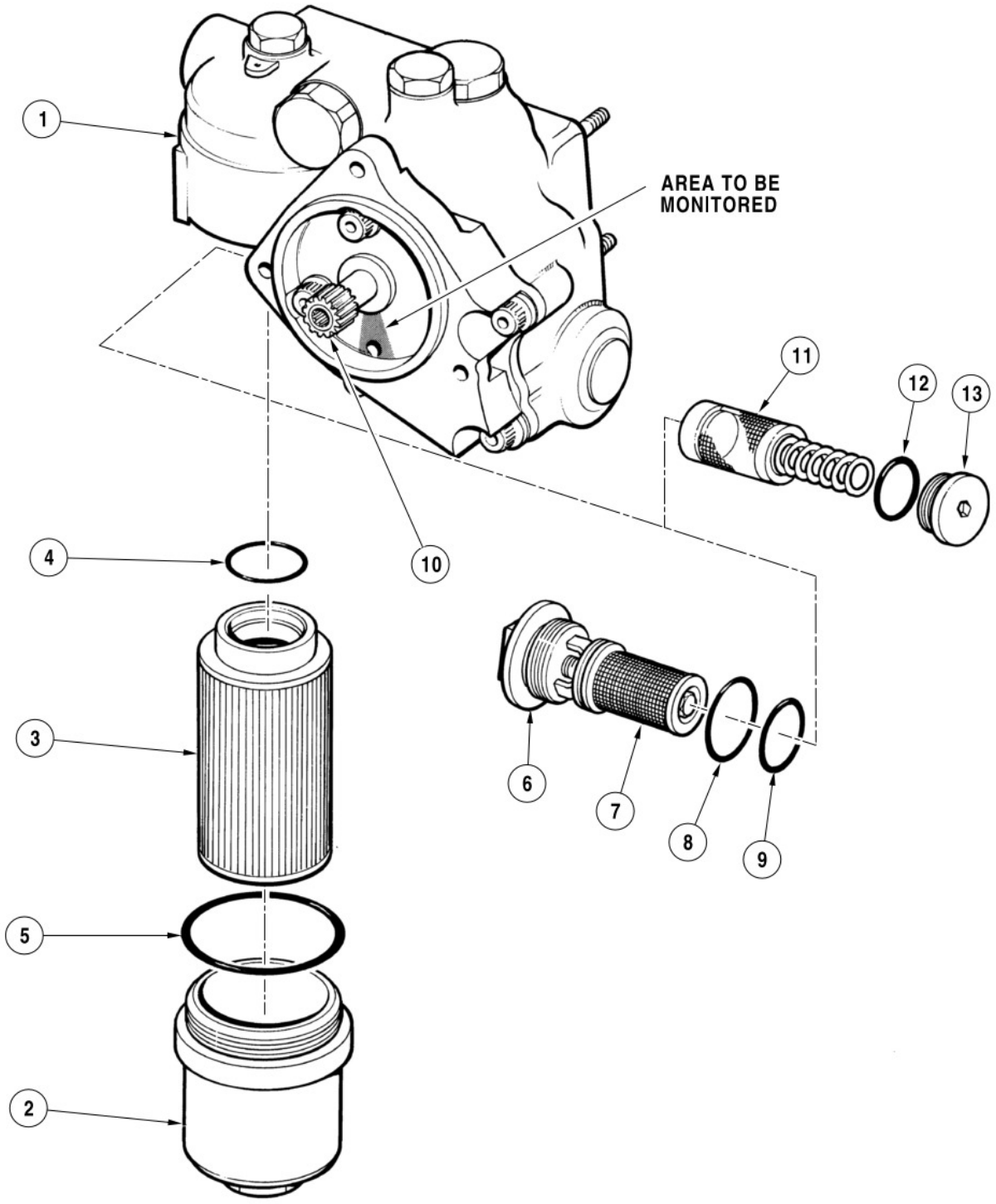
NOTE: The fuel pump can be either Sundstrand or Argo-Tech manufacture (Ref. IPC).

A. Removal of Fuel Pump Filters (Ref. Fig. 201)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 201 Removal/Installation of Fuel Pump Filters

(SHEET 1 OF 2)

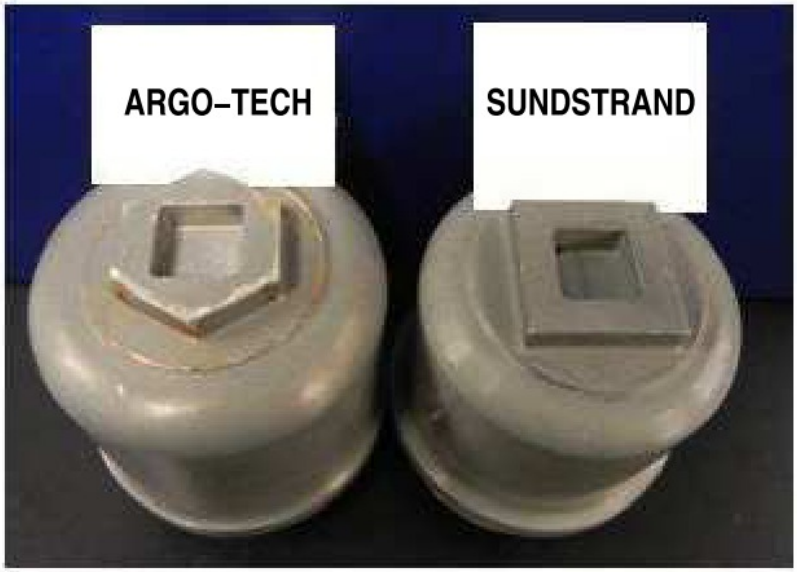


c227931

(SHEET 2 OF 2)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement





c231852

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Fuel Pump (Sundstrand or Argo-Tech)
2. Discharge Filter Cover
3. Discharge Filter Element
4. Preformed Packing
5. Preformed Packing
6. Inlet Filter Cover (Sundstrand Pump Assembly)
7. Inlet Filter Element (Sundstrand Pump Assembly)
8. Preformed Packing (Sundstrand Pump Assembly)
9. Preformed Packing (Sundstrand Pump Assembly)
10. Drive Shaft
11. Inlet Filter Assembly (Argo-Tech Pump Assembly)
12. Preformed Packing (Argo-Tech Pump Assembly)
13. Inlet Filter Cover (Argo-Tech Pump Assembly)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(1) Remove the discharge filter cover (2) from the fuel pump (1). Withdraw the filter element (3) and preformed packing (4) and discard the packing. At the life limit, withdraw the element and packing and discard both items (Ref. 72-00-00, INSPECTION, Periodic Inspection).

CAUTION: THE DISCHARGE FILTER ELEMENT IS DISPOSABLE AND MUST BE DISCARDED. DO NOT CLEAN.

(2) Remove the preformed packing (5) from the cover (2) and discard.

(3) Remove Sundstrand pump inlet filter:

(a) Remove the inlet filter cover (6) with the attached filter element (7) from the fuel pump (1). Do not disassemble the parts.

(b) Remove and discard the preformed packings (8 and 9) from the inlet cover (6).

(4) Remove Argo-Tech pump inlet filter:

(a) Using a $^{3}/_{8}$ inch Allen wrench, remove the inlet filter cover (13) with the attached filter assembly (11). Do not disassemble the parts.

(b) Remove and discard the preformed packing (12) from the inlet cover (13).

B. Installation of Fuel Pump Inlet Filters (Ref. Fig. 201)

(1) Install Sundstrand pump inlet filter as follows:

(a) Make sure the inlet filter element (7) is clean.

(b) Apply sufficient engine oil (PWC03-001) on new preformed packings (8) and (9) to provide surface film protection from abrasion, pinching or cutting damage during installation and assembly. Make sure the packings are completely wet.

(c) Assemble new preformed packings (8) and (9) under the head of the inlet filter cover (6) and in the groove of the inlet filter element (7) respectively.

(d) Put the assembled inlet filter element and cover (6) into the housing of the fuel pump (1) and turn the cover into the housing of the fuel pump until the inlet filter cover is fully seated on the pump housing.

NOTE: Make sure that the cover is fully seated with no visible preformed packing is seen in between the cover and fuel pump housing.

(e) Torque the cover 180 to 220 lbf.in. and lockwire.

(2) Install Argo-Tech pump inlet filter as follows:

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: WHEN INSTALLING THE INLET SCREEN ASSEMBLY INTO THE ARGO-TECH PUMP HOUSING ASSEMBLY, MAKE SURE THE SCREEN SEAT CONTACTS THE SEAT MACHINED IN THE HOUSING PRIOR TO COMPRESSING THE SPRING AND ENGAGING THE INLET SCREEN COVER THREADS. VERIFY PROPER INSTALLATION BY VISUALLY DETERMINING THAT THE UPPER EDGE OF THE SCREEN IS BELOW THE THREADS MACHINED IN THE PUMP HOUSING ASSEMBLY. IF IT IS NOT INSTALLED PROPERLY, THE SCREEN COULD GET HUNG UP ON THE SCREEN CENTERING LUGS AND THE UPPER EDGE OF THE SCREEN WOULD PROTRUDE INTO THE THREADED PORTION OF THE HOUSING.

(a) Install the inlet filter assembly (11) into the pump housing (1).

(b) Apply sufficient engine oil (PWC03-001) on new preformed packings (12) to provide surface film protection from abrasion, pinching or cutting damage during installation and assembly. Make sure the packings are completely wet.

(c) Install new preformed packing (12) in the groove on the inlet filter cover (13).

(d) Put the cover (13) over the spring, and turn the cover into the housing of the fuel pump (1). Turn the cover (13) until the preformed packing (12) touches the housing of the fuel pump (1) and continue the turning of the cover (13) until cover packing (12) is no longer visible and cover is bottomed against the fuel pump housing.

NOTE: Make sure that the cover is fully seated with no visible preformed packing is seen in between the cover and fuel pump housing.

(3) Torque the cover 30 to 50 lbf.in.

C. Installation of Fuel Pump Discharge Filter (Ref. Fig. 201)

CAUTION: DO NOT MIX THE SUNDSTRAND AND ARGO-TECH FUEL PUMP COVERS, BECAUSE ACCIDENTAL MIXING OF THE COVERS CAN RESULT IN BREAKING OF THE OUTLET FILTER WHILE ASSEMBLY (REF. FIG. 201).

NOTE: 1. Two service kits are available for the discharge filter. One kit covers consumable parts needed during discharge filter replacement; the other at discharge filter visual inspection (Ref. IPC).

NOTE: 2. Care must be taken, when you put first few threads of the filter cover (2) into the housing of the fuel pump (1). If you find unusual difficulty to turn the filter cover threads into the threads of the housing, do not continue the installation as seizure of both parts may occur. Remove the filter cover, inspect the filter cover threads and/or main housing threads and correct as necessary.

(1) Apply sufficient engine oil (PWC03-001) on the new preformed packing (5) and install it in the seal groove around the discharge filter cover (2).

(2) Apply sufficient engine oil (PWC03-001) on the new preformed packing (4) and install it in the seal groove of the filter element (3).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

CAUTION: DO NOT INSTALL THE FILTER ELEMENT AND COVER ASSEMBLY ONTO SPIGOT OF THE INTERNAL PUMP TO PREVENT FILTER DAMAGE AND CAN CAUSE UN-FILTERED FUEL TO ENTER INTO THE SYSTEM.

(3) Install the filter element (3) assembly into the fuel pump with the open end of the filter aligned with the spigot center of the pump and push upwards into the fuel pump (1) until filter (3) is bottomed.

NOTE: If the open end of the filter (3) is not aligned with internal spigot, filter element (3) will fall out while you release. If the filter element (3) falls during installation attempt, inspect for damage and/or cleanliness. Replace the filter element if it is damaged and/or dirty. Assemble and install again, do Steps (1) thru (3) .

(4) Install the discharge filter cover (2) over the filter element (3) and engage the threads in the housing of the fuel pump (1).

(5) Turn the cover (2) until the preformed packing (5) touches the housing of the fuel pump (1). Continue turning the cover (2) until cover packing (5) is no longer visible and cover is bottomed against the fuel pump housing.

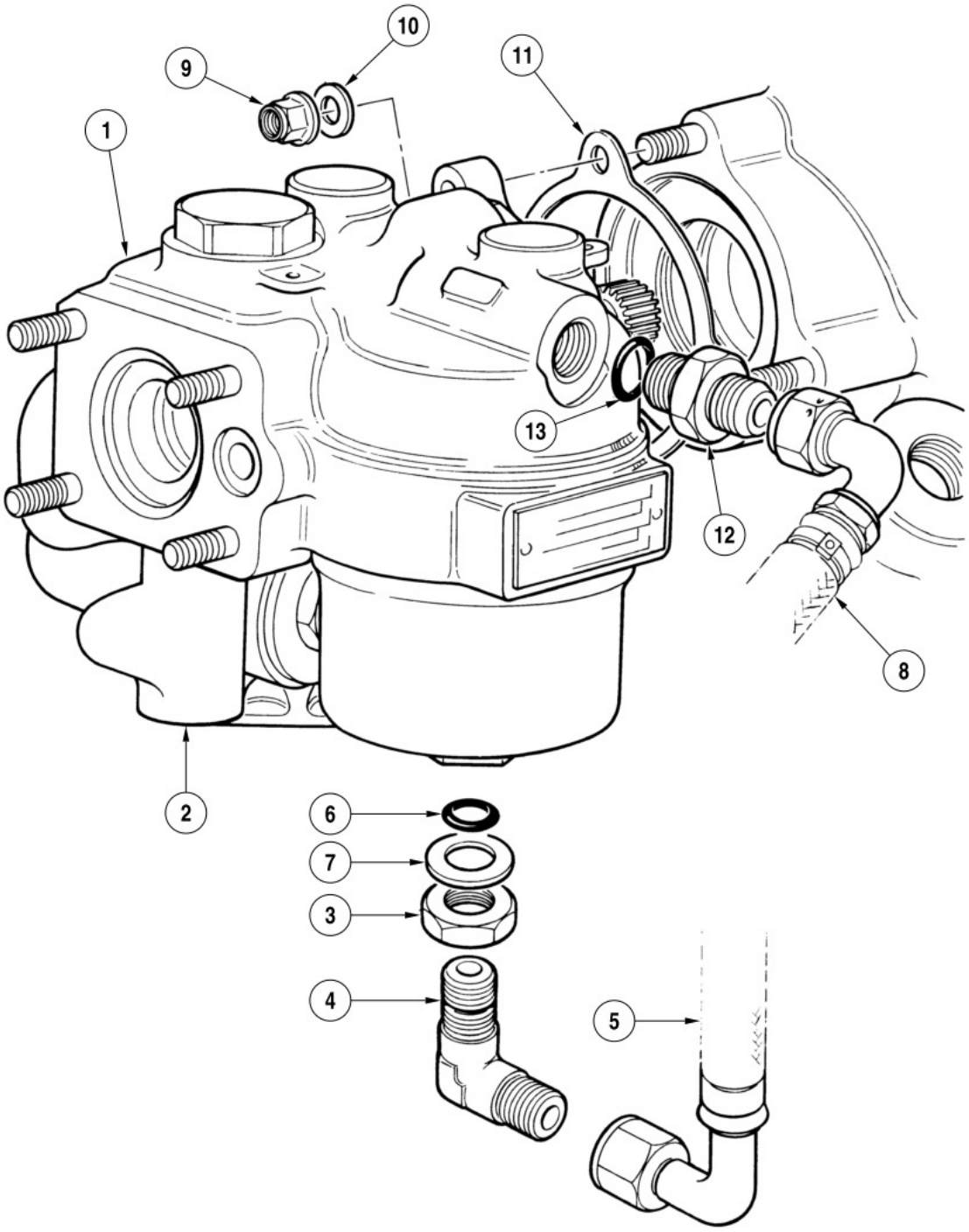
NOTE: Make sure that the cover is fully seated with no visible preformed packing and/or preformed packing is not come out of the cover and fuel pump housing.

(6) Torque the cover (2) 180 to 220 lbf.in. and lockwire.

6. Removal/Installation of Fuel Pump

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 202 Removal/Installation of Fuel Pump



c6967

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Fuel Pump
2. Seal Seepage Drain Port
3. Locknut
4. Elbow
5. Fuel Inlet Hose (from heater)
6. Preformed Packing
7. Back-up Ring
8. Fuel Outlet Hose (to FCU)
9. Self-locking Nut
10. Washer
11. Gasket
12. Nipple
13. Preformed Packing

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

A. Removal of Fuel Pump (Ref. Fig. 202)

(1) Remove the fuel control unit (Ref. 73-20-00).

(2) If the fuel pump (1) is being removed with the engine installed, disconnect the line from the airframe-supplied fitting installed in the seal drain port (2). Suitably cap the line connection and pump fitting.

(3) Disconnect the fuel hose (5) from the elbow (4) on the pump.

(4) Disconnect and remove the fuel outlet interconnecting hose (8) and suitably cap the hose connections and straight nipple (12) in the fuel pump (1).

(5) Support the fuel pump (1) and remove the self-locking nuts (9) and washers (10) that secure the pump to the accessory gearbox boss. Remove the pump and gasket (11).

(6) If the fuel pump is being replaced, remove the following parts for installation on the replacement pump:

(a) Remove the straight nipple (12) and preformed packing (13) from the outlet port of fuel pump (1). Install the blanking plug in the outlet port.

(b) Remove the elbow (4) and locknut (3) from the bottom of the pump. Discard the preformed packing (6) and back-up ring (7).

(c) Remove the airframe-supplied fitting and preformed packing from the drain port (2) of the fuel pump (1) and install the blanking plug in the drain port. If the fitting is of the elbow type, note the angular position before removal.

B. Installation of Fuel Pump (Ref. Fig. 202)

(1) On the Pre-SB1316 pump, make sure the drive coupling (10, Fig. 201) is secure on the pump internal splines by lightly pulling and pushing by hand. The coupling should be slightly loose, but should not move in or out. Check the protrusion of the coupling from the mounting face of the pump. Measurement should be 1.000 to 1.120 inches. Make sure the coupling is not displaced during assembly to the gearbox.

CAUTION: FOR Pre-SB1316 ENGINES: MAKE SURE DRIVE COUPLING (10, FIG. 201) IS CORRECTLY SECURED ON PUMP INTERNAL SPLINES BY LIGHTLY PULLING AND PUSHING BY HAND. COUPLING SHOULD BE SLIGHTLY LOOSE, BUT SHOULD NOT MOVE IN OR OUT. DO NOT REMOVE COUPLING FROM PUMP. CHECK PROTRUSION OF COUPLING FROM MOUNTING FACE OF PUMP. MEASUREMENT SHOULD BE 1.00 TO 1.12 INCHES. MAKE SURE COUPLING IS NOT DISPLACED DURING ASSEMBLY TO GEARBOX.

CAUTION: FOR Post-SB1316 ENGINES: PUMP COUPLINGS HAVE A STOP FLANGE AND MORE POSITIVE RETENTION. THERE SHOULD BE NO AXIAL MOVEMENT.

CAUTION: ALL TRACES OF GREASE MUST BE REMOVED FROM THE PUMP COUPLING SPLINES. ONLY ENGINE OIL IS TO BE USED AS A LUBRICANT. MAKE SURE THE OIL MIST LUBRICATION HOLE IS CLEAR.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(2) If a new fuel pump (1) is installed, do an engine depreservation (Ref. Chap. 72-00-00, SERVICING):

NOTE: Depreservation is not required if removed fuel pump is being re-installed.

(a) Remove the blanking plugs from the inlet and outlet ports of the fuel pump (1).

(b) Connect the supply of filtered fuel (PWC01-001) at the inlet pressure of 5 to 15 psig to the inlet port of the pump.

(c) Using a suitable turning device, rotate the drive shaft of the pump for approximately one minute until clear fuel is ejected from the outlet port.

(d) Remove the rotation device, disconnect the fuel supply and blank off the inlet port.

(e) Install a preformed packing (13) on the straight nipple (12) and install the nipple in the fuel outlet port. Tighten the nipple 110 to 120 lb.in. Install a suitable blanking cap.

(f) Install a preformed packing on the airframe-supplied fitting retained from the previous pump and install the fitting in the drain port (2) of the fuel pump (1). Tighten the fitting to the value recommended in the appropriate Aircraft Maintenance Manual. If the fitting is of the elbow-type, position at the same angle noted during removal (Ref. 70-00-00). Install a suitable blanking cap.

(g) Install the elbow (4) with a locknut (3), back-up ring (7) and preformed packing (6) in the boss in the bottom of the pump (Ref. 70-00-00, REMOVAL/INSTALLATION). Do not final torque the locknut until the angle has been established per Step (5) following.

CAUTION: FAILURE TO COMPLY WITH STEP (3) REQUIREMENTS OR LUBRICATION OF SPLINE WITH OTHER THAN ENGINE OIL MAY RESULT IN ACCELERATED SPLINE WEAR AND POSSIBLE FUEL PUMP DECOUPLE.

(3) Remove all traces of grease from the gearshaft bore and pump coupling. Make sure that the oil mist lubrication hole in the gearshaft restrictor is clear by manually inserting a No. 60 (0.040) drill into the restrictor hole, and turn the drill by hand. Make sure the teflon insert is installed, then liberally lubricate the splines with engine oil (PWC03-001).

NOTE: Operators may find it more convenient to install the FCU and fuel pump together as a unit after initial assembly of the two units on the bench. Refer to Chap. 73-20-00 for fuel control unit.

(4) Place the gasket (11) over studs of the accessory gearbox boss, then carefully position the fuel pump (1) over the studs. As the pump is pushed on, carefully engage the splines of the driveshaft with the corresponding splines in the gearshaft in the accessory gearbox. Secure the pump to the gearbox with washers (10) and self-locking nuts (9). Tighten the nuts 75 to 85 lb.in.

(5) Connect the fuel hose (5) to the elbow (4) on the pump. Final torque the locknut (3), as indicated in Step (2)(g) preceding, 145 to 155 lb.in. (Ref. 70-00-00, Table 401). Tighten the coupling nut of the hose 150 to 250 lb.in. Fasten the coupling nut and locknut with lockwire.

(6) Remove the blanking caps from the connection on the fuel outlet interconnecting hose (8) and from the straight nipple (12). Connect the hose to the nipple, but do not tighten until the

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

fuel control unit is installed; refer to Step (9), following.

(7) If the installation of the fuel pump (1) has been done with the engine installed:

(a) Remove the blanking caps from the seal drain line and from the airframe-supplied fitting in the drain port (2) of the fuel pump (1).

(b) Connect the line to the drain fitting and tighten to the torque value recommended in the appropriate Aircraft Maintenance Manual.

(8) Install fuel control unit (Ref. 73-20-00) if not installed together with fuel pump (Ref. Step (3)).

(9) Make sure the hose that interconnects the fuel pump outlet with the fuel control unit inlet is installed without twisting; tighten the connections at each end 270 to 300 lb.in. (Ref. 73-10-03).

7. Cleaning/Painting

A. General

(1) Clean the exterior surface of the fuel pump with petroleum solvent (PWC11-027) or (PWC11-031). To prevent the cleaning agent from entering the unit, do not remove the blanking caps and/or plugs.

(2) Wipe the surfaces dry with clean, lint-free cloths and/or compressed air to remove all residual solvent and contamination.

B. Inlet Filter Element

(1) Supply the petroleum solvent (PWC11-027) or (PWC11-031) through a 10-micron filter at 30 psig, to the outside of the screen mesh. Rotate the element during cleaning to make sure all areas of the mesh screen are flushed.

NOTE: Contaminants will usually collect inside the screen mesh at the cone section where the attachment bolt head is located.

(2) Reverse flush the inlet filter element with petroleum solvent to make sure all contamination is removed from the screen mesh.

(3) Remove the residual contaminants and dry the screen mesh using a blast of dry, clean, low-pressure air.

(4) Examine the inlet filter element for cleanliness and, if necessary, repeat the cleaning and reverse flushing operations until satisfactory.

8. Inspection/Check

A. General

(1) Inspect the mounting lugs visually for cracks and other defects. Cracks are not permissible; minor surface damage may be repaired.

(2) Inspect the machined mounting faces for gouges, nicks, scores, scratches, and similar surface defects which can damage the gasket between the pump and accessory gearbox and/or prevent proper mating of the fuel control unit to the pump. Particular attention

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

should be paid to the seal recess at the bypass port between the studs in mounting face for the fuel control unit.

(3) Inspect the bore of the inlet port for damage on the sealing surface.

(4) Check the studs for security and for damage to the threads.

(5) Inspect the seal seepage drain port and fuel outlet port adapter for damage to the threads.

(6) Check for leakage from seal drain port:

(a) Coupling drive oil leaks from fuel pump drain should not exceed 3 cc per hour. If limits are exceeded, replace plain seal on accessory gearbox pad (Ref. 72-60-00, Removal/Installation). If leaks are still present and exceed limits, ship unit for overhaul.

(b) Fuel leaks should not exceed 20 cc per hour. If leakage exceeds limit, ship fuel pump for overhaul.

NOTE: If limit is exceeded, inspect FCU driveshaft bearing area for fuel contamination (Ref. 73-20-00, Inspection/Check).

(7) Inspect the splines on the drive coupling for nicks, gouges and chips.

(8) Examine the external surfaces of the pump for corrosion and for the general condition of the anodic treatment.

(9) Check for signs of leakage from the joint between the pump cover and pump housing. If leakage is evident, return the pump for overhaul.

B. Filter Elements

NOTE: Refer to Paras. 5.A. and B. for removal and installation of fuel pump filters.

(1) Examine the inlet filter element for damage and/or breaks in the screen mesh.

(2) The discharge filter is disposable and should be subjected only to visual examination for defects if the life of the part has not expired.

C. Flexible Hoses

(1) Check that flexible hoses are fitted to avoid interference with the operation of the reversing linkage mechanism (Ref. 76-10-00).

D. Sundstrand Fuel Pump Coupling In-Situ Inspection

(1) Remove the drain line and fitting from the fuel pump drain port (2, Fig. 202).

CAUTION: IF A COTTON SWAB OR SIMILAR DEVICE IS USED FOR THIS INSPECTION, CARE MUST BE TAKEN TO MAKE SURE NO FOREIGN MATTER REMAINS IN THE DRAIN PORT OR ASSOCIATED FITTINGS AND LINES.

(2) Insert a cotton swab, or equivalent, 1.0 to 1.5 in. deep into the drain port of the fuel pump (Ref. Fig. 203).

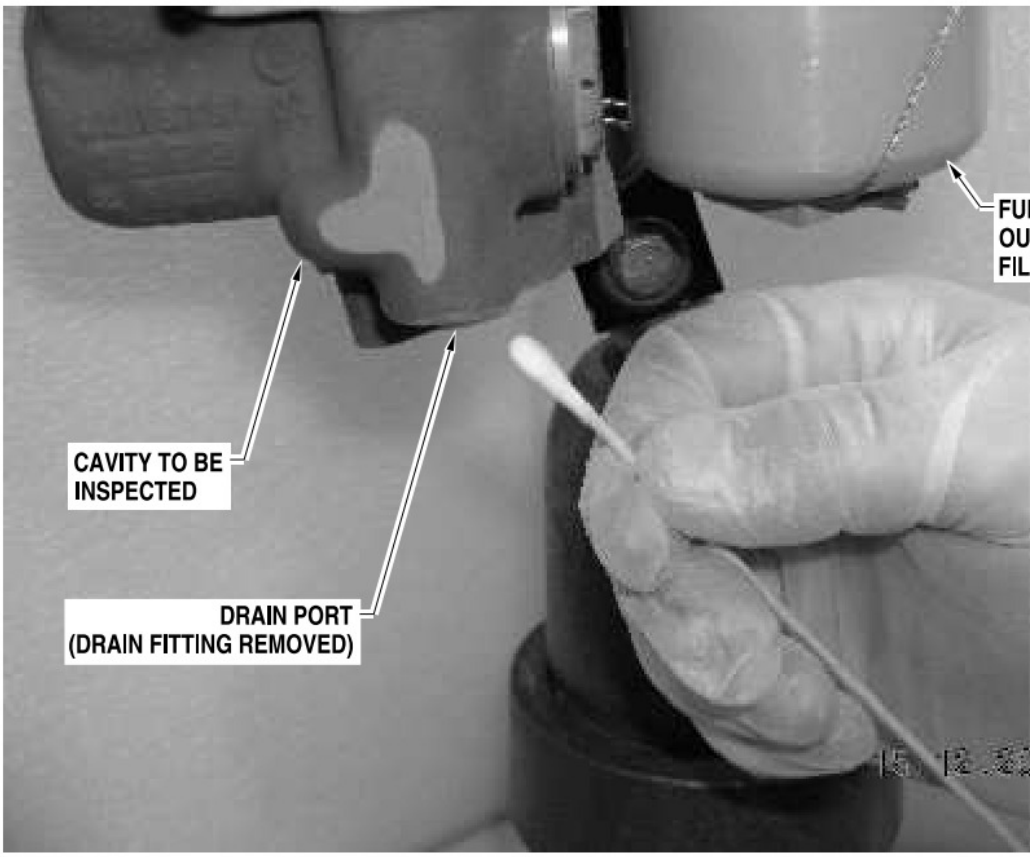
(3) Angle the cotton swab (Ref. Fig. 203) and roll it inside the drain port to collect evidence of a possible reddish-brown deposit (iron oxide).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(4) If no reddish-brown (iron oxide) stain is evident, the fuel pump may remain in service.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 203 Fuel Pump In-situ Inspection



c107468

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(5) If a reddish-brown (iron oxide) stain is evident:

(a) Remove fuel pump (Ref. Para. 6.).

(b) Examine the input coupling shaft area of the pump face (Ref. Fig. 201) for residue from fretting corrosion (iron oxide deposit). If none is present, the pump may remain in service.

(c) Evidence of corrosion residue indicates possible spline wear. Return the fuel pump to an approved overhaul facility.

(d) Install a replacement fuel pump (Ref. Para. 6.).

9. Approved Repairs

A. General

NOTE: After minor repairs are done, make sure that all areas are cleaned and corrosion preventive treatment is applied to aluminum surfaces. Do not allow filings to enter the unit.

(1) Minor damage such as burrs, scores, scratches, nicks, and similar surface defects may be cleaned up by blending with a fine stone and/or crocus cloth (PWC05-061), making sure that all sharp edges and high spots are removed.

(2) Clean up minor thread damage in the seal seepage drain port, and in the fuel inlet and outlet ports (when the nipple and elbow are removed) using a suitable swiss file or chase the threads using a suitable tap (Ref. IPC).

(3) Clean up minor thread damage on studs with a suitable swiss file, or chase threads using a 1/4-28 UNF-3A die.

(4) Blend out minor chips and nicks in the edges of splines, using a hard Arkansas stone, to remove rough edges and high spots that could damage mating splines and prevent proper meshing.

B. Corrosion Removal

(1) Light surface corrosion may be removed from the pump housing and cover using one of the two methods outlined in following text. Severe corrosion, indicated by surface etching, is cause for rejection of the pump, which must be returned to an overhaul facility.

(a) Remove corrosion deposits by vapor blasting with No. 1200 grit (PWC05-168) or finer. Make sure that areas adjacent to the corrosion are masked to prevent damage to the surrounding parts.

(b) Remove corrosion deposits by local polishing with No. 44 grit abrasive cloth (PWC05-101), followed by crocus cloth (PWC05-061). Remove as little base metal as possible.

(c) Apply anti-corrosion treatment to the repaired area (Ref. Subpara. C.).

C. Application of Anodize Surface Treatment

(1) The anodic anti-corrosion finish on the pump housing and cover may be repaired by a local application of chemical treatment solution (PWC05-166). This treatment must be used on

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

areas where minor repairs have been carried out by blending and/or polishing. To apply the treatment, proceed as follows:

(a) Prepare the surface by cleaning with a cleaning solution (PWC11-038). Keep the surface wetted with the cleaning agent for one to five minutes.

(b) Rinse with clean water to remove all traces of the cleaning agent.

WARNING: ALODINE SOLUTION IS EXTREMELY DANGEROUS. IT CONTAINS AN OXIDIZING INGREDIENT WHICH CAN CAUSE AN EXPLOSION IF IT COMES IN CONTACT WITH COMBUSTIBLE MATERIALS SUCH AS PAINTS AND SOLVENTS.

WARNING: DURING THE USE OF ALODINE SOLUTION AVOID BREATHING VAPORS; USE ONLY WITH ADEQUATE VENTILATION. WEAR CHEMICAL SAFETY GOGGLES AND RUBBER GLOVES WHEN HANDLING. IN CASE OF SKIN CONTACT, WASH WITH SOAP AND WATER; IF IRRITATION APPEARS AND PERSISTS, CONSULT A PHYSICIAN (MEDICAL DOCTOR). IN CASE OF EYE CONTACT, FLUSH EXTENSIVELY WITH FRESH WATER AND CONSULT A PHYSICIAN IMMEDIATELY.

(c) Apply chemical treatment solution (PWC05-166) to prepared surface by brush, swab or spray method and allow three to five minutes setting time.

(d) Rinse with clean water to remove all traces of excess chemical treatment solution and allow surface to air dry.

(e) Examine treated surface and make sure that repair area is completely covered. Reapply treatment as necessary.

10. Adjustment/Test

Do a functional check of the fuel pump during the next engine test run (Ref. 71-00-00).

11. Fault Isolation

For detailed procedures refer to Chapter 72-00-00, FAULT ISOLATION.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement